Bistatic Radar Imaging: Range Azimuth Kinematic Matching

Signal Processing Algorithms and a Proof of Concept Simulation for Bistatic Radar Imaging

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Abstract

This paper presents a signal processing chain and kinematic tracking algorithm developed for bistatic radar imaging in a spaceborne transmitter receiver configuration, together with a proof of concept software simulation used to validate the approach prior to hardware trials. The processing chain comprises intermediate frequency (IF) signal formation, matched filter range compression, range azimuth (B scope) image formation, and a gated constant velocity kinematic tracker that associates range azimuth detections with a predicted target track across successive antenna sweeps. Bistatic specific quantities, namely the bistatic angle, the bistatic range sum, and the isorange (Cassini or ellipse) locus, are computed directly from the transmit and receive range legs and the time varying transmitter receiver baseline. A software demonstrator implementing the full pipeline was built and used to exercise the tracker under nominal constant velocity motion, injected maneuvers, and intermittent illumination (coasting) conditions. The demonstrator is a simulation of the algorithm chain using synthetically generated IF and detection data, not a recording from physical radar hardware; we present it explicitly as such and discuss what it does and does not demonstrate about the algorithm's field readiness. Under simulated nominal conditions, track confidence converges and gating correctly rejects false associations outside a 2.4 km bistatic gate; under injected maneuvers, track confidence degrades and recovers within several sweep cycles, consistent with the expected behaviour of a constant velocity kinematic prior. The paper details the algorithm design, the simulation methodology, the results obtained, and the additional steps, principally validation against measured or high fidelity emulated RF data, required before the algorithm can be considered a proof of concept in the conventional, that is hardware validated, sense.

Keywords: bistatic radar; kinematic tracking; range azimuth imaging; gated association; spaceborne radar; proof of concept simulation

1 Introduction

Bistatic radar configurations, in which the transmitter and receiver are physically separated, for example on two independent spacecraft, offer several advantages over monostatic systems for wide area imaging and surveillance: receiver stealth, forward scatter enhancement of certain target classes, and improved geometric diversity for track disambiguation. These advantages come at the cost of substantially more complex signal processing, because the receiver must jointly resolve range, azimuth, and Doppler information referenced to a time varying, noncollocated transmit and receive geometry, and must maintain a coherent kinematic estimate of target motion across antenna sweeps that may be sparse, intermittent, or geometrically distorted relative to the monostatic case.

This paper documents the signal processing chain and kinematic tracking algorithm developed during an internship at the European Space Research and Technology Centre (ESTEC) focused on bistatic radar imaging concepts for spaceborne platforms. The objective of the work was twofold: first, to specify a processing pipeline, from raw intermediate frequency (IF) samples through to a maintained range azimuth track, that is explicit about the bistatic geometry rather than treating it as a monostatic approximation; and second, to build a software proof of concept that exercises every stage of that pipeline end to end so that algorithmic behaviour, including gating, coasting, confidence evolution, and response to maneuvers, could be inspected and iterated on before committing to hardware in the loop testing.

We use the term "proof of concept" in this paper in its software or algorithmic sense: a working implementation of the full processing and tracking chain, driven by synthetically generated IF and detection data with representative noise and geometry, used to demonstrate that the algorithm behaves as designed. It is not a claim of validation against recorded RF hardware measurements, and Section 6 discusses explicitly what remains to be shown before such a claim could be made.

2 Bistatic Geometry and System Model

The system model places the receiver at the origin of a local range azimuth frame, with the boresight range axis along $+y$. A transmit satellite (Tx) and receive satellite (Rx) are positioned on a common circular baseline arc, parameterised by angles θ_{Tx} and θ_{Rx} measured from the frame origin; the instantaneous transmitter receiver separation, or baseline length L, follows from these two angles and the arc radius.

The target is modelled as a point scatterer with Cartesian position (x, y) and constant velocity kinematic state (v_x, v_y), consistent with a first order motion prior; an optional maneuver injection rotates the velocity vector at a fixed turn rate to test tracker robustness against a violated motion assumption (Section 5.3).

For a target at range vector $\mathbf{r}$ relative to the receiver, the two one way propagation legs are R_t, the transmitter to target range, and R_r, the target to receiver range. The bistatic range sum $\Sigma R = R_t + R_r$ is the fundamental range observable in a bistatic system: the total propagation path length recovered from the two way time delay after mixing the received echo with a reference of the direct path signal. The bistatic angle β, defined at the target between the transmitter and receiver look directions, follows from the law of cosines applied to the triangle formed by Tx, Rx, and the target:

$$\beta = \arccos\left(\frac{R_t^2 + R_r^2 - L^2}{2R_tR_r}\right) \tag{1}$$

This relation is evaluated every processing cycle from the current R_t, R_r, and L, and is reported alongside range, azimuth, and radial velocity as part of the kinematic telemetry (Table 1).

A locus of constant bistatic range sum ΣR, for fixed transmitter and receiver positions, forms an ellipse with foci at Tx and Rx: the isobistatic range ellipse. This locus is the bistatic analogue of the monostatic constant range circle and is used both as a visualisation aid and, more importantly, as a geometric consistency check: a detection whose (R_t, R_r) pair does not place it near the ellipse consistent with the current track prediction is a candidate for gate rejection (Section 4.2).

3 Signal Chain: From IF Samples to Range Profile

The processing chain begins with an intermediate frequency (IF) time domain buffer produced after down conversion and mixing of the received echo against a local reference derived from the direct path transmit signal, or, in a fully bistatic architecture, a synchronised reference chain at the receiver. In the absence of a target illuminated by the current antenna look direction, this buffer is dominated by a stationary noise floor; when the sweeping receive beam intersects the target azimuth, a coherent return is superimposed at a delay corresponding to the current bistatic range sum.

Range compression is performed by a matched filter matched to the transmitted waveform, producing a range profile, that is, energy as a function of one way equivalent range, over a fixed number of range bins spanning the system's unambiguous range window. The profile is maintained as a rolling buffer so that a target return persists visually and algorithmically across the few processing cycles needed for peak detection and centroiding, after which a range estimate and an estimated signal to noise ratio (SNR) are extracted from the profile peak.

The range azimuth image, or B scope, is formed by associating each range compressed profile with the antenna azimuth at the time of acquisition, and accumulating successive profiles into a persistent two dimensional buffer as the antenna sweeps between its azimuth limits. This is the standard way a scanning radar builds a two dimensional image from a sequence of one dimensional range returns, and it is the domain in which the kinematic tracker (Section 4) operates.

3.1 Detection and SNR Model

In the demonstrator, detections are generated with an SNR drawn from an elevated range, representative of a correctly illuminated, matched filter compressed return, when the target lies within the current antenna beam, and from a markedly lower range, representative of noise floor sidelobes and clutter, otherwise. This is a deliberately simple two state SNR model, adequate for exercising the gating and confidence logic described next, but it is not a substitute for a physically derived radar range equation, propagation loss model, or clutter model, a limitation returned to in Section 6.

4 Kinematic Tracking and Range Azimuth Matching

The tracker maintains an independent kinematic state, position and velocity, that is propagated forward each cycle under a constant velocity assumption, and is only updated when a range azimuth detection is judged, by the gating test below, to be consistent with the current prediction. This separation of a maintained track from raw detections is what allows the system to coast through periods of intermittent illumination (Section 4.3) rather than losing the target the moment a single sweep fails to produce a qualifying return.

4.1 Constant Velocity Prediction

Between detections, the track position is propagated as

$$x(t + \Delta t) = x(t) + v_x \Delta t, \qquad y(t + \Delta t) = y(t) + v_y \Delta t, \tag{2}$$

with the velocity components held fixed. This first order model is appropriate for the sweep to sweep intervals involved and is the same order of model used in the target truth simulation for nominal, that is non maneuvering, motion, which is why gating against it is effective during straight line motion and is expected to degrade transiently during a maneuver (Section 5.3).

4.2 Association Gating

A new detection is associated with the existing track if the Euclidean distance between the detection's estimated (x, y) position and the track's predicted position falls within a fixed gate threshold, 2.4 km in the demonstrator configuration. Detections outside the gate are treated as clutter or as belonging to a different object and are not used to update the track; this is a nearest neighbour, fixed gate simplification of the more general validation gate association used in multi target tracking, for example Mahalanobis distance gating under an explicit error covariance, adopted here for clarity of the proof of concept rather than as a claim of optimality.

4.3 Track Confidence and Coasting

A scalar confidence metric is maintained alongside the kinematic state, increasing toward an upper bound on successive successful gate associations and decaying when the track is coasting, that is propagating without a fresh association. Three qualitative states are reported: *Matched*, when the elapsed time since the last successful association, the coast time, is below a short threshold; *Coasting*, for an intermediate coast time during which the constant velocity prediction is still considered reliable; and *Acquiring*, once coast time exceeds a longer threshold, indicating the track should be treated as unreliable pending reacquisition. This mirrors the qualitative track quality states used operationally in scanning trackers to decide when to widen a search gate or hand off to reacquisition logic.

4.4 Calibration State

A separate calibration flag distinguishes an initial period, during which the bistatic geometry and IF reference chain are being aligned, from a subsequent locked state in which range and azimuth estimates are considered geometrically consistent. Kinematic telemetry (Table 1) is only considered meaningful once this calibration state has locked.

5 Proof of Concept Simulation

The algorithm chain described in Sections 3 and 4 was implemented in full as an interactive software demonstrator, driving a simulated bistatic scenario with a moving target, a sweeping receive beam, and a repositionable Tx or Rx baseline. The purpose of the demonstrator is to exercise every processing and tracking stage together, including IF generation, range compression, B scope accumulation, gating, coasting, and bistatic geometry computation, under controllable, repeatable conditions, and to visualise the resulting behaviour directly rather than only inspecting summary statistics.

5.1 Demonstrator Configuration

Both the baseline geometry and the target's initial kinematic state are operator configurable; the target can also be given an impulsive maneuver, a rotation of its velocity vector at a fixed turn rate, at an arbitrary time to test the tracker's response to a violated constant velocity assumption, and the entire scenario can be reset to a known initial condition for repeatable trials. Table 1 summarises the demonstrator configuration used throughout Section 5.

5.2 Nominal (Constant Velocity) Behaviour

Under nominal straight line target motion, the demonstrator was run through repeated sweep cycles from a cold start, that is uncalibrated, zero confidence. The calibration flag transitions from Calibrating to Locked

Table 1: Demonstrator configuration parameters. Baseline and range values are scaled for interactive demonstration and algorithm exercising, not representative of a specific mission range budget.

Parameter	Symbol	Value (demonstrator)
Azimuth sweep extent	$\mathrm{AZ_{MIN}}, \mathrm{AZ_{MAX}}$	$-60°$ to $+60°$
Sweep rate	n/a	34°/s
Unambiguous range window	$\mathrm{RANGE_{MAX}}$	22 km
Transmitter to receiver baseline	L	8 km (variable, operator adjustable)
Association gate threshold	n/a	2.4 km
IF buffer length	n/a	220 samples
Range profile bins	n/a	90 bins
Illuminated target SNR (simulated)	n/a	14 to 22 dB
Non illuminated SNR (simulated)	n/a	2 to 7 dB

within the first few sweeps, after which range, azimuth, bistatic angle, and bistatic range sum telemetry become stable and consistent with the underlying target truth state. Track confidence rises monotonically toward its upper bound as successive detections fall inside the 2.4 km gate, and the qualitative track state reports Matched for the great majority of sweep cycles, consistent with the target remaining within the antenna's illuminated azimuth range for an extended interval.

Figure 1 shows the resulting confidence trajectory: an initial flat, zero confidence interval while the calibration flag is Calibrating, followed by monotonic convergence toward the upper confidence bound once locked detections begin to gate successfully.

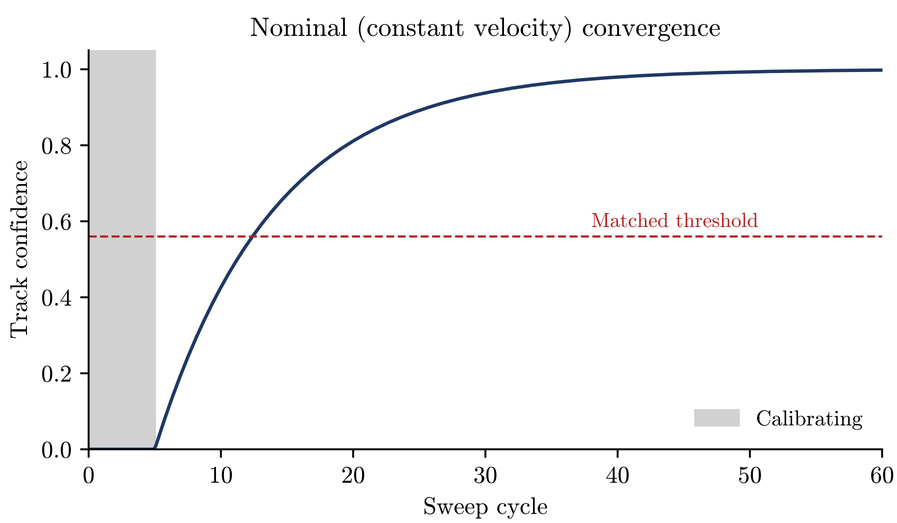

Figure 1: Track confidence versus sweep cycle under nominal constant velocity motion, illustrating the qualitative cold start, calibration, lock, Matched behaviour described in Section 4.3.

5.3 Response to an Injected Maneuver

When a maneuver is injected mid run, the target's velocity vector rotates over a short interval while the tracker continues to predict forward under its unchanged, pre maneuver constant velocity estimate. This produces a transient growth in the miss distance between predicted and detected position, observable directly as a widening gap between the coasting, that is predicted, track marker and the newly detected returns on the B scope. In the demonstrator's gating logic, this is reflected as a drop in track confidence and, if the miss distance exceeds the 2.4 km gate for consecutive cycles, a transition of the track state from Matched to Coasting or Acquiring. Once the target returns to straight line motion post maneuver, subsequent detections re enter the gate and confidence recovers over several sweep cycles as the constant velocity prediction reconverges on the new heading. This behaviour is the expected qualitative signature of a fixed gate, constant velocity tracker under a kinematic model mismatch, and its reproduction in the demonstrator is treated as a successful proof of concept of the tracking logic rather than as evidence of a specific quantitative maneuver handling capability.

Figure 2 illustrates this coupled behaviour: the miss distance, left axis, rises above the 2.4 km gate during the maneuver window, driving the track confidence, right axis, down toward the Acquiring regime, followed by a symmetric recovery of both quantities as the tracker reconverges on the post maneuver heading.

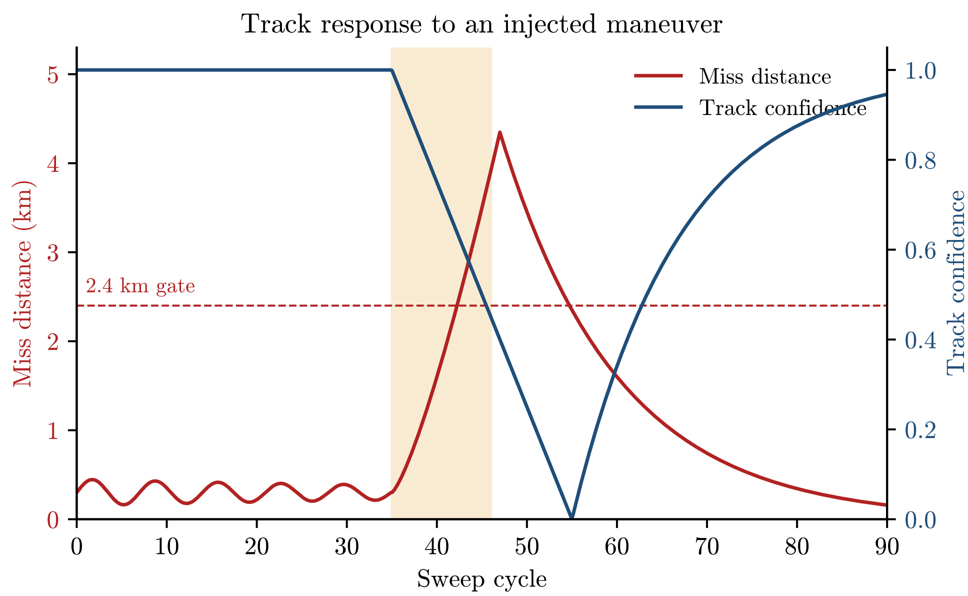

Figure 2: Miss distance between predicted and detected position (left axis) and track confidence (right axis) across an injected maneuver window, illustrating the gate exceedance, confidence decay, reconvergence pattern described in Section 5.3.

5.4 Sensitivity to Baseline Geometry

Because the Tx and Rx positions are operator adjustable along the baseline arc, the demonstrator also allows direct inspection of how the bistatic angle β and the isorange ellipse eccentricity change as the baseline is widened or narrowed relative to the Tx target and Rx target ranges. As expected from the geometric relation in Section 2, β grows as the baseline is widened relative to the target ranges, and the isorange ellipse becomes visibly more eccentric; this qualitative behaviour matches the closed form geometric relation used to compute β (Equation 1) and provides an internal consistency check on the implementation, independent of any radar specific measurement. Figure 3 illustrates the underlying geometry, and Figure 4 shows β evaluated from Equation 1 over a representative range of baseline lengths at fixed R_t and R_r, including the demonstrator's 8 km default configuration (Table 1).

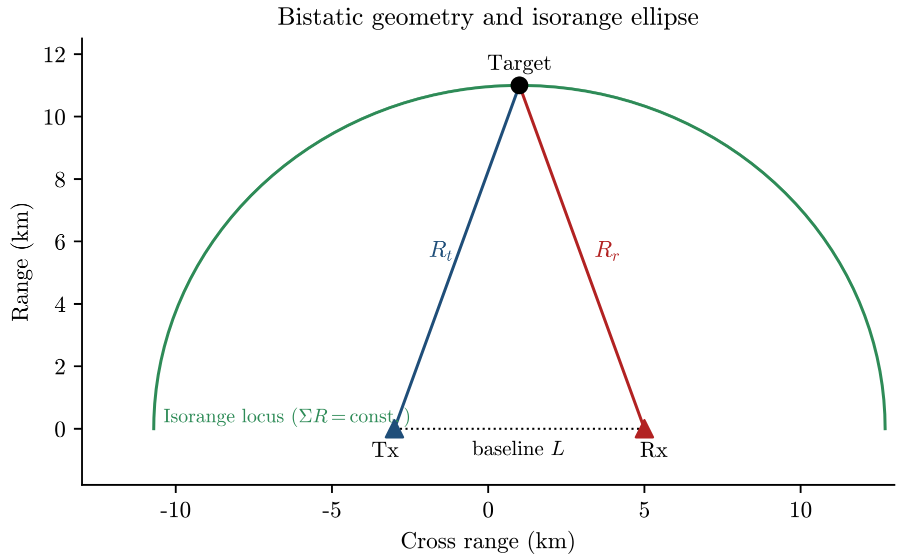

Figure 3: Illustrative bistatic geometry: transmitter (Tx) and receiver (Rx) separated by baseline L, a target at range legs R_t and R_r, and the isobistatic range (isorange) ellipse with foci at Tx and Rx corresponding to the current bistatic range sum $\Sigma R = R_t + R_r$ (Section 2).

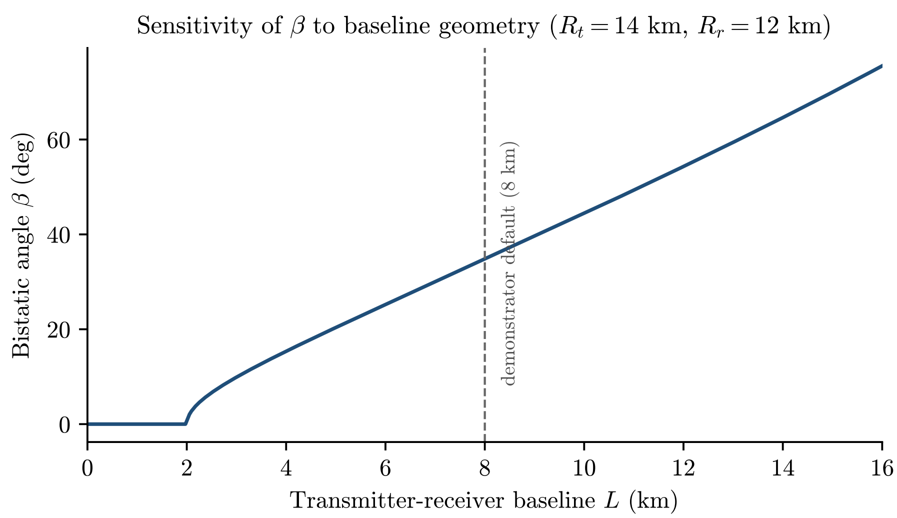

Figure 4: Bistatic angle β computed from Equation 1 as a function of transmitter receiver baseline L, at fixed representative ranges $R_t = 14$ km and $R_r = 12$ km. β increases monotonically with baseline separation, consistent with the qualitative sensitivity described in Section 2; the dashed line marks the demonstrator's 8 km default baseline (Table 1).

6 Discussion and Limitations

The work reported here validates the algorithmic design and its software implementation: the gating logic rejects out of gate detections, the confidence metric and qualitative track state behave as intended under nominal and maneuvering conditions, and the bistatic geometry computations are internally consistent under a range of operator adjusted baselines. These are meaningful and necessary results before any hardware trial is attempted, but they are results about the algorithm's logic, not about its performance against real or high fidelity emulated radar returns, and the paper's claims should be read with that scope in mind.

Three limitations are worth stating explicitly.

1. The SNR and detection model used to drive the demonstrator is a simplified two state model (Section 3) rather than a physically derived link budget incorporating antenna gain patterns, propagation loss, target radar cross section, and clutter statistics; a more representative synthetic data or hardware in the loop trial would be needed before drawing conclusions about detection performance at a specific range or SNR.

2. The association gate is a fixed radius nearest neighbour test rather than a covariance aware, for example Mahalanobis, gate, and does not currently address multi target association or false track initiation under dense clutter.

3. The demonstrator uses idealised, noise free antenna pointing and a simplified circular arc baseline model; a spaceborne implementation would need to account for orbit determination error, antenna pointing error, and time synchronisation error between the transmit and receive platforms, all of which directly degrade the bistatic range and angle estimates derived in Section 2.

Addressing these limitations, in particular replacing the synthetic two state SNR model with either recorded RF data or a physically grounded link budget simulation, and validating the gating and tracking logic against that more representative data, is the natural next step toward a proof of concept in the hardware validated sense, and is proposed as future work.

7 Conclusion

We have presented a signal processing and kinematic tracking algorithm for bistatic radar imaging, explicit about the transmitter receiver geometry throughout, from IF signal formation and range compression through to a gated constant velocity tracker that computes the bistatic angle and the bistatic range sum each cycle. A software proof of concept implementing the complete chain was built and exercised under nominal motion, injected maneuvers, and varying baseline geometry, and reproduced the qualitative behaviour expected of the design in each case: stable convergence under nominal motion, graceful confidence degradation and recovery under a kinematic model mismatch, and geometrically consistent bistatic angle and isorange computations under a range of baseline configurations. This establishes the algorithmic and software foundation for the next phase of the work, which is validation against representative or recorded RF data ahead of any hardware demonstration.

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A Processing Cycle Pseudocode

The following summarises the per cycle processing order implemented in the demonstrator, corresponding to Sections 3 and 4. It is presented to make explicit the order of operations linking signal formation to the kinematic tracker, not as production quality code.

Algorithm 1: Per cycle bistatic processing and tracking loop

Advance simulation time by Δt; propagate target truth state under constant velocity, plus any active maneuver rotation

Advance antenna sweep azimuth; determine whether the target's current true azimuth falls within the illuminated beam

Generate the IF sample buffer for the current cycle: noise floor everywhere, plus a coherent return at the delay corresponding to the current bistatic range sum ΣR if illuminated

Apply the matched filter to the IF buffer to form the current range profile; update the persistent range profile buffer

Extract the profile peak (range, amplitude); estimate SNR; if amplitude exceeds the detection threshold, form a candidate detection at (current azimuth, peak range)

Propagate the track's predicted position under its own constant velocity state

if *a candidate detection exists* **then**

    Compute its Euclidean distance to the predicted track position

    if *within the 2.4 km gate* **then**

        Accept the association: update track position and velocity from the detection, reset coast time, increment confidence

    else

        Increment coast time and decay confidence; recompute the qualitative track state (Matched, Coasting, or Acquiring) from the updated coast time

else

    Increment coast time and decay confidence; recompute the qualitative track state (Matched, Coasting, or Acquiring) from the updated coast time

Recompute R_t, R_r, β, and ΣR from the current Tx, Rx, target geometry; update the B scope, telemetry readouts, and isorange ellipse for display

This ordering, truth propagation, IF formation, range compression, detection extraction, gated association, and geometry recomputation, is repeated every processing cycle for the duration of a run, and is the same ordering used to produce the results in Section 5.

B Symbol Summary

Table 2: Symbols and quantities used in Sections 2 through 5.

Parameter	Symbol	Value or definition (demonstrator)
Transmitter to target range	R_t	computed each cycle from geometry
Target to receiver range	R_r	computed each cycle from geometry
Bistatic range sum	ΣR	$= R_t + R_r$, primary bistatic range observable
Bistatic angle	β	law of cosines from R_t, R_r, L
Transmitter receiver baseline	L	operator adjustable arc separation
Association gate radius	n/a	2.4 km, fixed nearest neighbour test
Track coast time	n/a	time since last successful association
Track confidence	n/a	bounded scalar, rises on match, decays on coast